

Energy Procurement Optimization

Multi-Objective Linear Programming (MOLP) Approach

Decision Tree

Linear Programming

Green Energy

Portfolio Optimization

Project Summary

Objective

A manufacturing plant must procure electricity every hour to meet production demand.

Determine the **optimal hourly portfolio mix** – how much electricity to procure from each source – to balance simultaneously:

- (1) **minimize total cost;**
- (2) **minimize supply risk;** and
- (3) **Minimizing CO₂ intensity.**

Context & Challenges

No single source satisfies all three sub-objectives. The optimal mix depends on prices, demand levels, and how much the company values sustainability.

Three available sources:

	Cost	Supply Risk	Carbon Cost
(1) Standard Contract	Medium	Yes	Upon grid mix
(2) RE Certified Contract	High	Yes	No
(3) Spot Market	Low (avg)	No	Upon hourly grid mix

Proposed Model

Scenario-based MOLP with:

- (1) **Decision tree** structures demand & price scenarios; and
- (2) **Multi-objective Linear Programming** solves optimal portfolio allocation in expectation across all scenarios.

Pipeline

1 – Decision Framework

Define scenarios & MOLP problem structure

2 – Data Collection

Obtain firm and electricity market data.

3 – Scenario Building

Estimate probability for decision trees from data.

4 – Solve & Optimize

LP solver per scenario to obtain optimal portfolio.

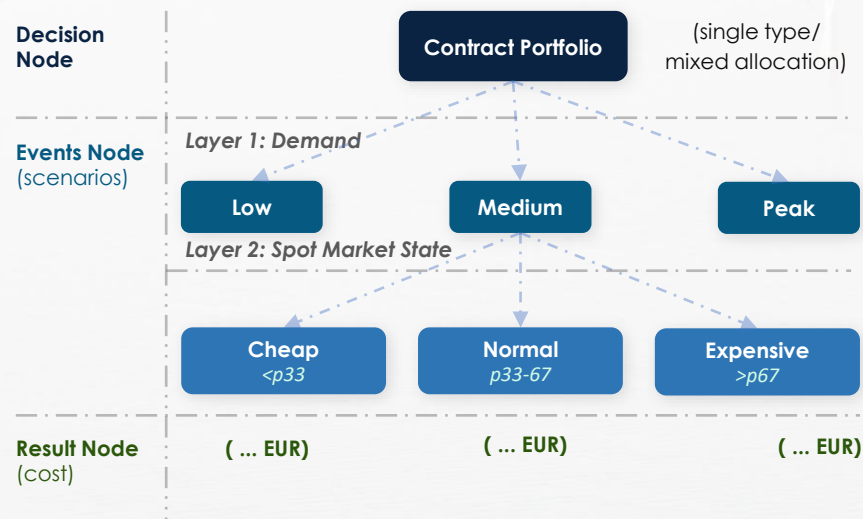
5 – Sensitivity Analysis

Across different weights/contract premium.

Decision Framework

① Decision Tree – Scenarios

- Probabilities of each scenario are estimated from data:
 - Demand tiers → Load_Type frequency in DAEWOO dataset
 - Price tiers → Percentile splits (P33 / P67) from ENTSO-E Finland spot prices
- Each leaf node probability = product of conditional probabilities along its path.
- Total scenarios:
 - If spot prices involved: up to 9 leaf nodes (3 × 3)
 - Fixed contracts only: 1 leaf node



② MOLP – Optimization Model

$$\min_{x_1(t), x_2(t), x_3(t)} \sum_s p(s) \cdot [w_1 \cdot \text{Electricity Cost}(s) + w_2 \cdot \text{Supply Risk Loss}(s) + w_3 \cdot \text{Carbon Cost}(s)]$$

Note: t = each hour in planning horizon (8,760 hrs/year)

$x_1(t)$ $x_2(t)$ $x_3(t)$ **Decision Variables (solved by LP)** – MWh bought from Standard/ RE-Certified Contract/ Spot

w_1 w_2 w_3 **Priority Weights (set by the company)** – How much company values each sub-objective

$s, p(s)$ **Scenario s & probability $p(s)$ of each scenario**

③ 4 Company Profiles (Solve model for each type of company)

	Balanced	Cost – focused	Supply Risk – averse	Green-focused
w_1	0.34	0.7	0.2	0.2
w_2	0.33	0.2	0.7	0.1
w_3	0.33	0.1	0.1	0.7

Key Findings

01

Cost & Carbon are nearly insensitive to priorities

Finland's grid is already clean and inexpensive. The payoff table shows only 5% Cost spread and 21% Carbon spread across the 3 extreme strategies - even green-focused or risk-averse companies pay almost the same as cost-focused ones.

02

Supply Risk is where priorities truly diverge

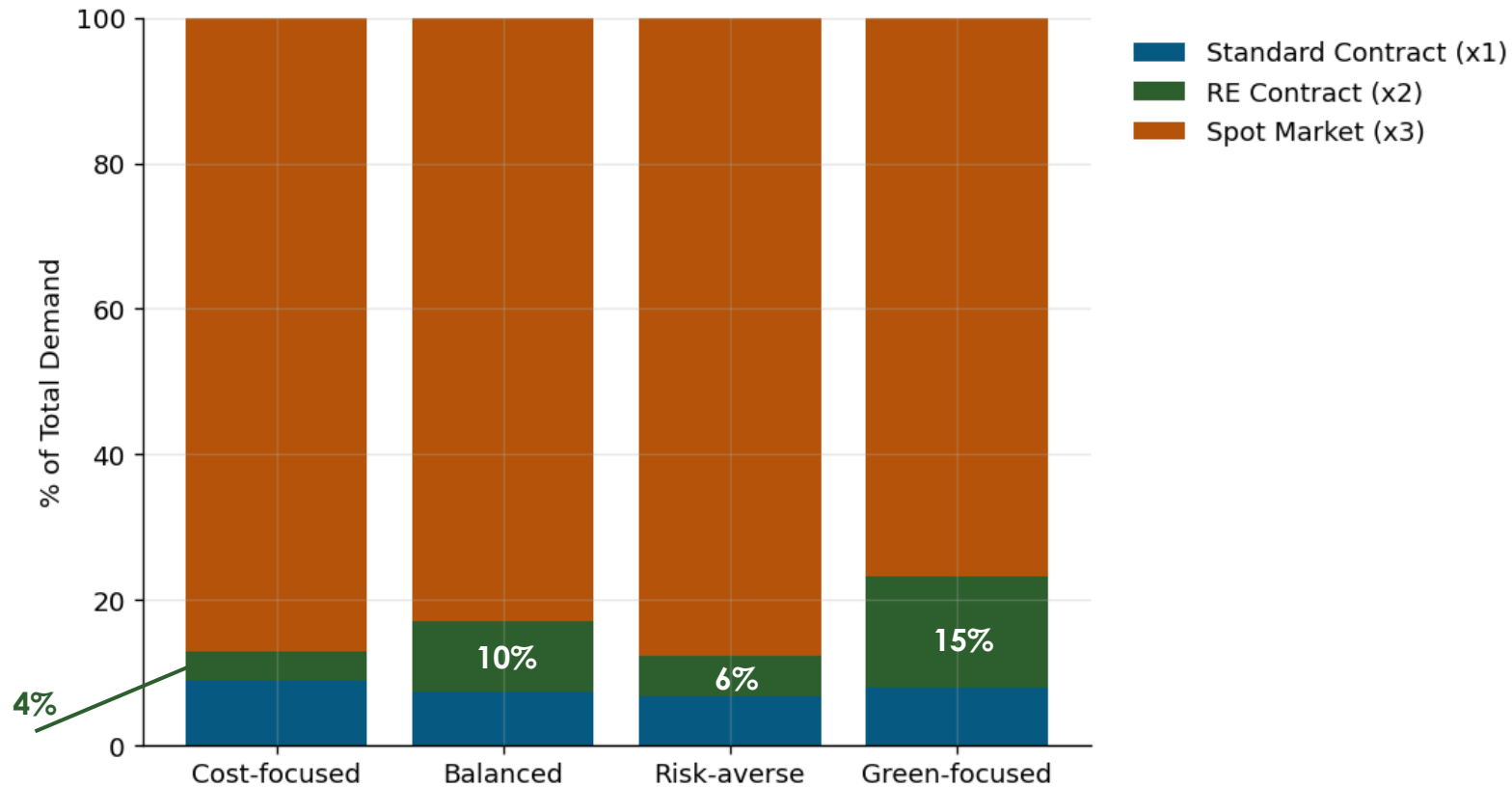
Risk-averse drives supply risk cost to exactly EUR0; Green-focused has the highest risk (~EUR7,200). Because both standard) and RE contracts carry the same generator-failure risk, chasing the green objective (more x2) increases risk cost as a side effect.

03

'Going green' is cheaper in electricity/carbon cost – but shifts risk cost

RE contract share roughly triples (4% -> 15%) from Cost-focused to Green-focused, at near-zero cost penalty. The real trade-off a company faces isn't financial or environmental - it's exposure to contract-supply risk.

Portfolio Mix by Company Profile



Key Implications

1. Spot dominates

75-87% across all profiles - Finland's spot market is cheap and reasonably clean by default

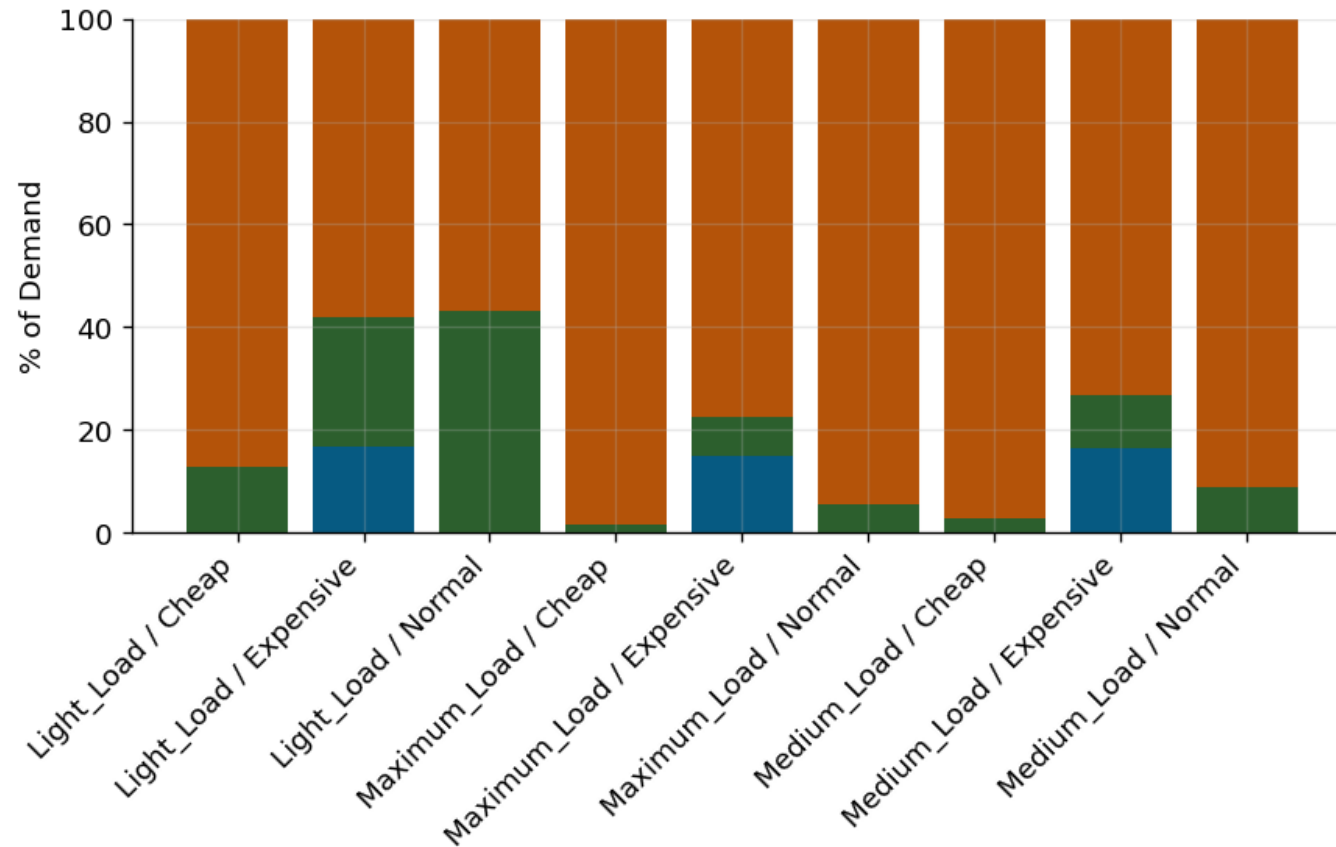
2. RE Contract triples

4% -> 15% from Cost-focused to Green-focused - the green objective targets x2 specifically

3. Standard contract stable

~7-9% across all profiles - not a major lever for any priority

Portfolio Mix of Balanced Profile



Key Implications

1. Cheap, low-demand hours

almost entirely spot
(Light Load/Cheap: 87% spot)

2. High-stakes hours

contract usage rises sharply
(Light Load/Normal: 43% RE
Light Load/Expensive: 42% combined contract)

-> the LP locks in price certainty exactly when it matters most.

- Standard Contract (x1)
- RE Contract (x2)
- Spot Market (x3)

Optimization Model – Objectives

Aggregate Objective Function

$$\min_{x_1(t), x_2(t), x_3(t)} \sum_s p(s) \cdot [w_1 \cdot \text{Electricity Cost}(s) + w_2 \cdot \text{Supply Risk}(s) + w_3 \cdot \text{Carbon Cost}(s)]$$

$x_1(t)$ $x_2(t)$ $x_3(t)$ **Decision Variables (solved by LP)** – MWh bought from Standard Contract/ RE-Certified Contract/ Spot Market

w_1 w_2 w_3 **Priority Weights (set by the company)** – How much company values cost savings/ supply reliability/ green score.

s $p(s)$ **scenario s** (combination of time context, demand level, spot price state) & **probability $p(s)$** of each scenario

$$\textcircled{1} \text{ Elec Cost} = \sum_t \sum_{i=1}^3 p_i(t) \cdot x_i(t)$$

where:

p_1 standard contract price = $E[\text{spot}] \times 1.10$
 p_2 RE contract price = $E[\text{spot}] \times 1.15$
 $p_3(t)$ spot price at hour t

Unit: (EUR)

Note: t = each hour in planning horizon (8,760 hrs/year)

$$\textcircled{2} \text{ Supply Risk} = \sum_t \text{short_fall}(t) \cdot [p_{\text{bal}} - p_3]$$

where:

short_fall (t) = $\epsilon \cdot (x_1 + x_2)$: possible amount that contract generator fails to deliver
 ϵ : fraction of committed amount contract generator fails to deliver
 p_{bal} : price to buy from balancing market

Captures loss from unmet demand (EUR)

Assume baseline $\epsilon(t) = 0.02$,

$$\textcircled{3} \text{ Carbon Cost} = \sum_t (x_1(t) + x_3(t)) \cdot ci(t) \cdot \text{carbon_price}$$

where:

ci(t): grid carbon intensity, varies by hour

Unit: EUR

Optimization Model – Constraints

The LP solves for $x_1(t)$, $x_2(t)$, $x_3(t)$ that **minimize the probability-weighted sum of all three objectives** across all defined scenarios.

Constraints:

	Inequality/Equation	Explanation
(1) Demand Coverage	$x_1(t) + x_2(t) + x_3(t) = D(t)$	Must cover demand at each hour
(2) Standard Contract Cap	$x_1(t) \leq \overline{C_1}$	Cannot exceed agreed contract capacity
(3) RE Contract Cap	$x_2(t) \leq \overline{C_2}$	Cannot exceed agreed contract capacity
(4) Non-negativity	$x_1(t), x_2(t), x_3(t) \geq 0$	Cannot procure negative quantities

* Note: Assume $\overline{C_1}, \overline{C_2} = 50\%, 20\%$ of average $D(t)$